A Weekly Demand Briefing for Scotland's Emergency Departments

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AI Mastery Capstone: Industry Integrated AI System Synthesis

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28 October 2026

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Overview and Industry Context

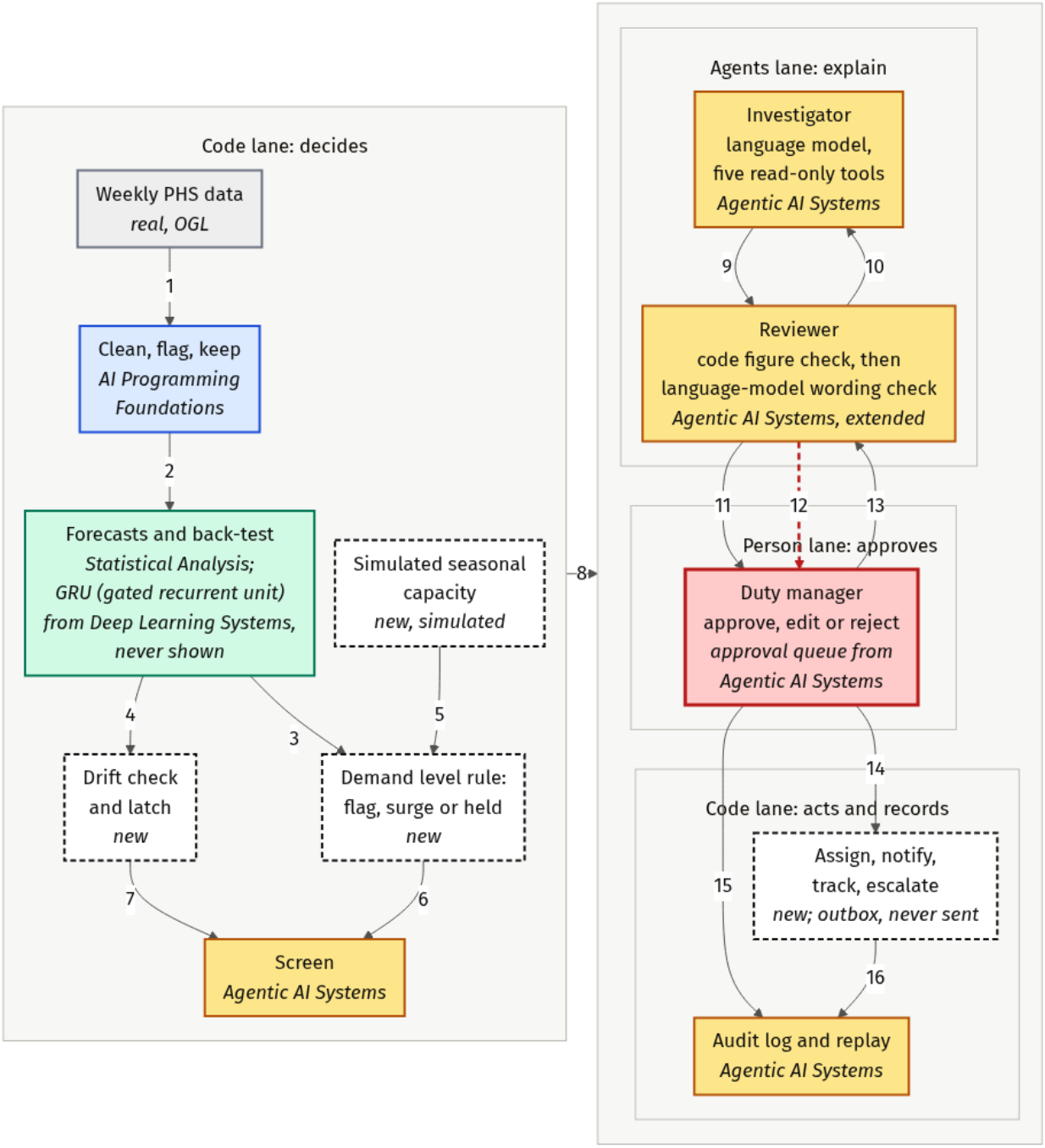
The industry is healthcare operations: unplanned attendances at Scotland's emergency departments (EDs), for the duty managers who staff the front door. England's National Health Service (NHS) rota code gives resident doctors their rosters at least 6 weeks before a rotation (NHS England, 2025), so an unusually busy week meets a front door staffed for the usual week. ED crowding is linked with higher morbidity and mortality (Scottish Government, 2016). A forecast one to four weeks ahead cannot reset a rota, but can inform extra shifts, cover from the staff bank or held leave.

The system forecasts attendances and writes a brief when a hospital's week looks unusually busy or its forecasts stop fitting. At this stage it gives the duty manager additional information beside their existing process: it replaces no current process and sets no staffing on its own. It needs shadow mode, a pilot with managers who did not build it, and tuning, before it could be a primary operational tool.

Why artificial intelligence (AI)? It is needed less for the forecast than for its explanation. Simple models forecast demand from open data on accident and emergency (A&E) attendances (Public Health Scotland [PHS], 2026), used under the Open Government Licence (OGL). A gated recurrent unit (GRU), a small neural network, runs beside them only as a comparator (Petravić et al., 2025). A duty manager needs each forecast explained, with its range and how far to trust it, so a language model agent drafts that explanation from read-only tools (Yao et al., 2023), while code sets every number (Figure 1).

Figure 1

Code Decides, the Agents Explain, a Person Approves

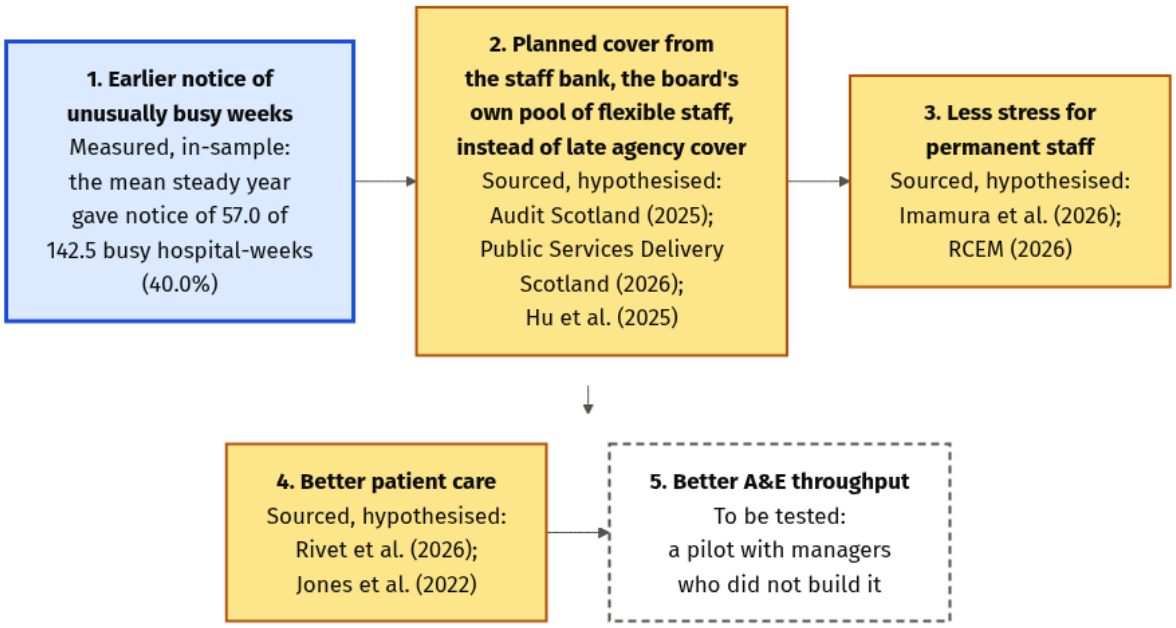


The value is in planned cover. In 2024/25 Scotland spent £256m on agency staff of all kinds and, separately, £353m on the nursing bank (Audit Scotland, 2025, p. 15). Any saving

would be mainly the agency premium, with wider value in staff welfare and patient care. Figure 2 shows the value chain.

Figure 2

The Value Chain, Each Link Labelled by Its Evidence



Note. Busy means 1.05 times simulated capacity or more; the mean steady year averages 2019 and 2023 to 2025. Sources: Audit Scotland (2025), Public Services Delivery Scotland (2026), Hu et al. (2025), Imamura et al. (2026), Royal College of Emergency Medicine (2026), shown as RCEM, Rivet et al. (2026) and Jones et al. (2022).

Integration Rationale

I integrated four prior capstone projects (Table 1): data analysis, statistics, deep learning and agentic systems, whose language model brings in generative AI. A fifth, Applied Machine Learning, shaped two decisions: results per site and the pilot.

Each taught a lesson the design keeps. AI Programming Foundations kept one unexplained merchant's payments, flagged, and the answer changed without them, so the COVID-19 weeks stay in, flagged, as a real shock. In Conduct a Statistical Analysis Using Python ("Statistical Analysis") the changes' standard deviation, 18.63 points, was below either period's spread (22.36 and 21.29), so pairing gave a more precise test. Here a paired interval compares the GRU with the primary model, the simple model chosen per series, on the same weeks. In Deep Learning Systems the best naive forecast placed the day's peak better than either network, so simple models stay the bar to beat. In Agentic AI Systems the Investigator cited figures no tool had returned and the check caught only one of ten, so here code writes every figure and a person approves every brief.

On five replay series at four horizons, outside weeks when forecasts had stopped fitting, that interval favoured the primary in 16 of 20 comparisons and the GRU in none. The one new technology, Anthropic's client library for the frontier model, sits behind Agentic AI Systems' model gateway and ran only after a local model failed a test stated in advance.

Table 1

Prior Capstone Projects and What Each Supplies

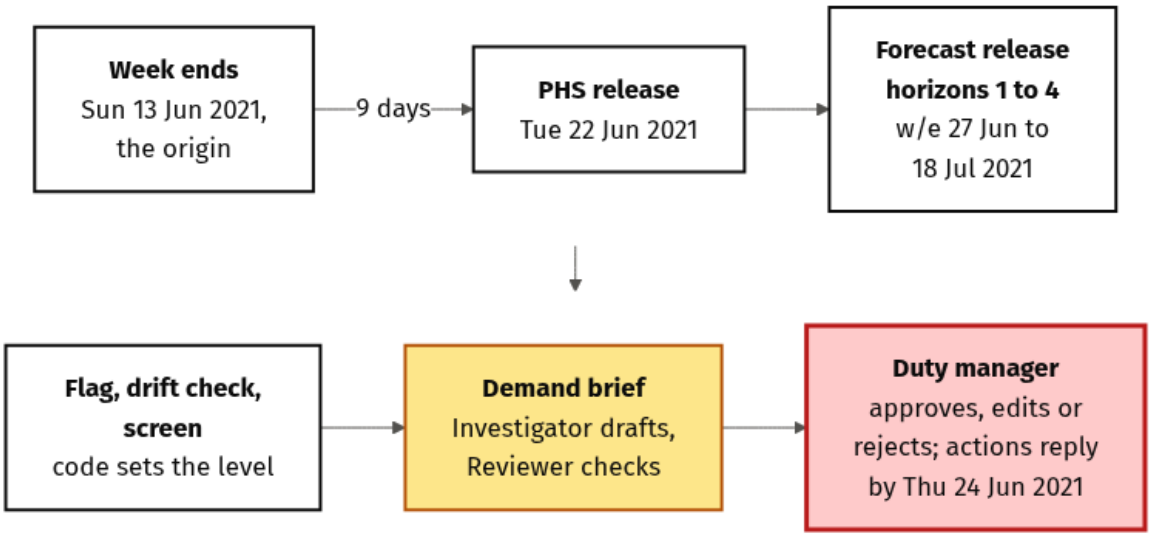
Prior capstone project	What it supplies here
AI Programming Foundations	The weekly load, the COVID-19 weeks kept and flagged
Statistical Analysis	The back-test and bootstrap (Hyndman & Athanasopoulos, 2021, Section 12.5) and the paired test of the GRU
Deep Learning Systems	Seasonal naive, the 4-week moving average and the GRU comparator
Agentic AI Systems	The screen, read-only tools, the figure check, replay, approval and audit

System Design and Technical Decisions

Figure 1 reads in arrow order, down the left column and then the right: code decides, the agents explain, the duty manager approves, edits or rejects each brief, and code then acts and records. The data are weekly unplanned A&E attendances at 30 hospitals and nationally, 593 weeks to the week ending (w/e) 11 October 2026. Each week is released 9 days after it ends, so the forecasts cover the four weeks after each release, its release horizons (Figure 3).

Figure 3

One Week, From the Tuesday Release to the Duty Manager's Decision



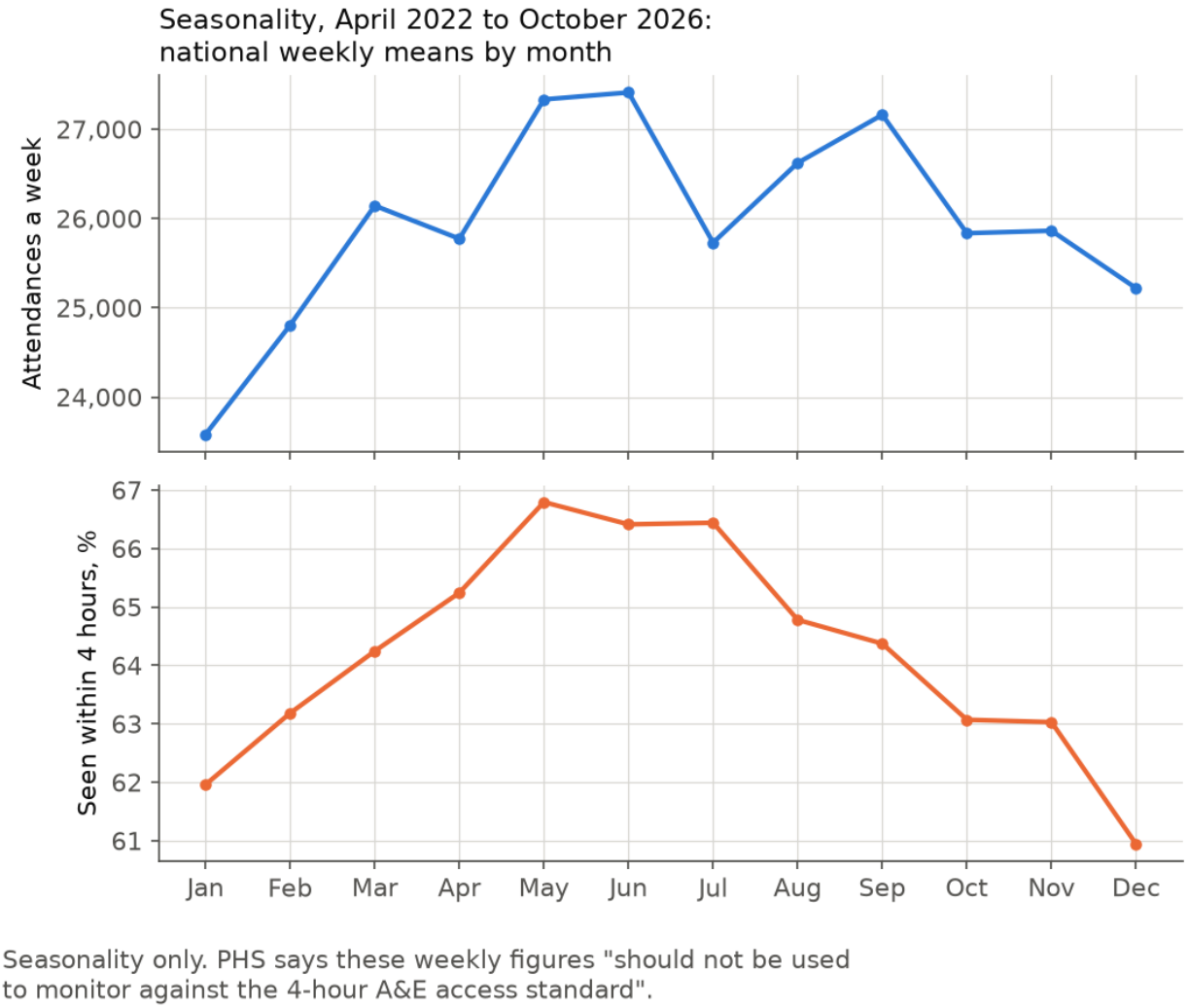
Note. Dates from Ninewells' demand week ending 13 June 2021, whose brief was rejected.

Four simple models forecast each series, among them seasonal naive, the same week last year (Hyndman & Athanasopoulos, 2021, Section 5.2). The open data hold no beds or staffing, so I use simulated capacity, a seasonal baseline of past demand: the 90th percentile of the hospital's attendances in the same weeks of the previous three years. Code flags an outlook, one hospital's next four weeks, when the primary model's last 104 errors give any week at least a

50% chance of reaching 1.05 times capacity. The demand level runs from 1 to 4, a flag raises a week to level 2, and its bands borrow only the structure of England's Operational Pressures Escalation Levels (OPEL) framework (NHS England, 2024). The level shows demand unusually high for the time of year, not pressure, need or priority: attendances peak in May and June, and 4-hour performance is worst in winter (Figure 4).

Figure 4

Monthly Attendances Against 4-Hour Performance



Note. Data from PHS (2026), the source of the quotation in the image.

The drift alarm fires when attendances fall outside the one-week forecast plus or minus 1.96 standard deviations of its recent errors (Hyndman & Athanasopoulos, 2021, Section 5.5) for two released weeks running. It sets the drift latch, a switch that marks the hospital's forecasts untrusted and stops its flag until reset. A surge, an upward breach, takes the latest released week's level. In the design only a person resets the latch; this build simulates each reset. At each reset the primary model is re-selected by its rolling-origin error on the latest 104 released weeks (Hyndman & Athanasopoulos, 2021, Section 5.10). Table 2 gives four tradeoffs.

Table 2

Four Tradeoffs That Shaped the Design

Choice	Gain	Cost
A simple primary model; the GRU only a comparator	A forecast checkable in one sentence	Any pattern the simple models miss
Capacity simulated from past demand	A level tested on real attendances	Illustrative, not about beds
A rare flag	In-sample, steady years: 53.8% of flags borne out, against 21.2% without one	Only 21.0% of busy outlooks flagged
Hosted Claude Sonnet 5.5, switchable to a local model in Ollama	5 of 7 briefs verified, against none locally	Cost per call, an outside service, no pinned version

The Investigator, as in Agentic AI Systems, reasons between tool calls in the ReAct pattern (Yao et al., 2023), writing each figure as a placeholder that code fills. The Reviewer is code first: it checks every figure against its tool result and screens for the words the Medicines and Healthcare products Regulatory Agency (MHRA) lists as indicative of a medical device (MHRA, 2023, p. 20). A language model then checks the wording. A brief that fails twice is marked "not verified, do not act" and can never be approved. After approval, code sets staffing actions, owners and deadlines, and a language model words each notification, kept in an outbox,

never sent. Each brief opens with what a duty manager needs at a glance and what to do, then the evidence (Figure 5).

Figure 5

A Brief as the Duty Manager Receives It

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Royal Infirmary of Edinburgh at Little France, week ending 23 Aug 2026 (figures released
  Tue 01 Sep 2026)

At a glance
- Level 2 of 4: demand may be higher than usual for the time of year from w/e 06 Sep
  2026 to w/e 27 Sep 2026. The level does not measure pressure, need or priority.
- Why: at least one of the next four weeks may run 5% or more above this hospital's
  high-week mark for the time of year. That mark is the weekly count 9 weeks in 10
  stayed under at this time of year over the last three years.
- Records show: the latest week, w/e 23 Aug 2026, had 2,740 attendances. A high week
  here at this time of year is about 2,540 a week.
- Expected attendances: the forecasts centre on about 2,460 to 2,660 a week; 8 weeks in
  10 should fall between about 2,400 and 2,850.
- The forecasts: the four methods' central figures differ by up to 8.0% in any one week.
- How accurate they have been: on past weeks these forecasts were out by 2.9% to 4.5% on
  average, for the first of the four weeks.
- Past flags here: when this hospital was flagged before, demand did turn out high, a
  week 5% or more above the mark, 4 times out of 5.
- No warning that the forecasts have stopped matching recent weeks.

What to do, if you approve
- Duty manager: Check whether extra shifts or bank staff are needed for w/e 06 Sep 2026
  to w/e 27 Sep 2026.
- Rota coordinator: Hold new leave requests for w/e 06 Sep 2026 to w/e 27 Sep 2026 until
  the duty manager's check is done.

The detail follows: each forecast with its ranges, past forecast errors and the track
  record (examples/current_brief_edinburgh.md).
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Note. The top of the Royal Infirmary of Edinburgh's brief for w/e 23 August 2026, written by code: the latest flag with no drift alarm in the five series the system runs. W/e 20 September 2026 ran 6% above the high-week mark (simulated capacity). The examples/ folder holds the full brief.

Ethical Considerations and Responsible AI

Automation bias and transparency. Agentic AI Systems named automation bias, approving a brief without weighing it (Goddard et al., 2012), as the next risk at approval. Against

it, every brief opens with the level's meaning, what the records show, the expected attendances with their likely range, or why no forecast can be trusted, and the track record. The approver, the duty manager, judges each against six criteria written in advance. A person can override every output, following human oversight as a principle, not a legal duty (European Union, 2024, Article 14).

Misuse and status as additional information. As Statistical Analysis warned its counts could be misused to name code breakers, this tool is never for ranking hospitals, justifying cuts or monitoring the 4-hour standard. It never sees patient data and informs staffing, not any patient's care, so on my reading of the MHRA's guidance it is not a medical device (MHRA, 2023).

Bias by hospital size and accountability. Seasonal naive's median error is 11.0% at the smallest sites and 5.4% at the largest. The share of outlooks raised varies by site more than by size, from 35.7% at the Royal Hospital for Children to none at Glasgow Royal Infirmary, both in the largest band. So, as Applied Machine Learning reported results per public body, each brief shows its own site's error, and the level is never read as need. Owners are by role, on the structure of OPEL's actions by level (NHS England, 2024), and a clinical safety reviewer would sign off before any live use.

Evaluation and Reflection

Every figure is in-sample: the design was tuned on the data that scores it. Error is the mean absolute percentage error (MAPE; Hyndman & Athanasopoulos, 2021, Section 5.8). Figure 6 scores the eight goals, set in advance.

Figure 6

Goals G1 to G8, Each Against Its Threshold

Goal	Met if	Result	Verdict
G1 Forecasts a manager can plan with	within 1 point at every release horizon	-0.18 to -0.74 points against the best simple model	**met**
G2 Honest uncertainty	90% to 98%	95.7% to 96.4% of actuals inside	**met**
G3 Shocks are caught	by Tue 31 Mar 2020; at most 1 a year	known Tue 31 Mar 2020; at most 1 alarm a year	**met**
G4 No unchecked number approved	every approved brief passes; no not-verified brief approved; plant caught	5 of 5 approved pass; 2 not verified, neither approved; plant caught	**met**
G5 The agent never sets a level	always the rule's level	7 of 7 briefs written	**holds**
G6 Briefs are useful to a person	6 of the 7 briefs approved with no factual change, and Reader B makes the approver's decision on at least 3 of the 4 they read	5 of 7 (the two demand-week briefs were refused and rejected); Reader B agreed on 4 of 4	**not met**
G7 Raised levels rare in steady years	under 10% in every steady year	over 10% in 3 years, 10.4% to 11.7%	**narrowly missed**
G8 A flag tells a manager something	at least twice the base rate	2.54 times the base rate, steady years	**met**

G1 to G3, G7 and G8 are in-sample, on 2015 to 2026. G4 to G6 are on the frontier set's seven replay briefs. G5 is an invariant, not a test.

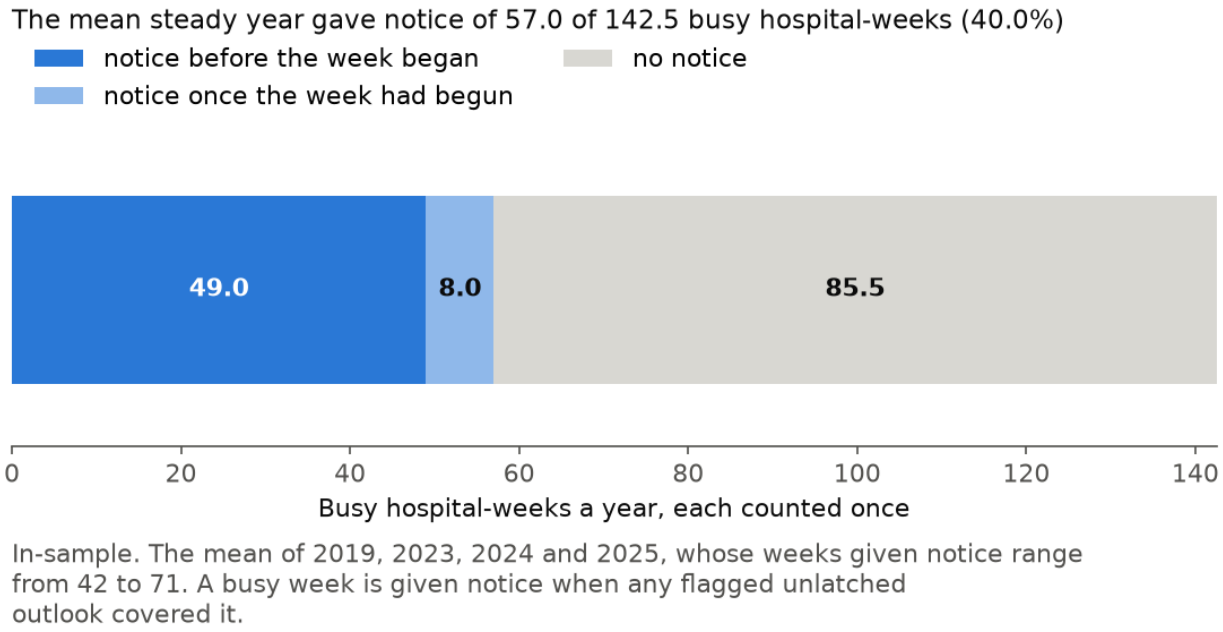
Note. Reader B: a second reader who did not build the system.

What worked. Five goals are met, G1 to G4 and G8, and "the agent never sets a level" (G5) holds. For G1, the primary model's national MAPE outside shock weeks is within 1 point of

the best simple model at every release horizon, and measurably lower only at the fourth. For G8, the flag alone gave notice of 57.0 of 142.5 busy hospital-weeks in the mean steady year (Figure 7).

Figure 7

Busy Hospital-Weeks Given Notice in the Mean Steady Year



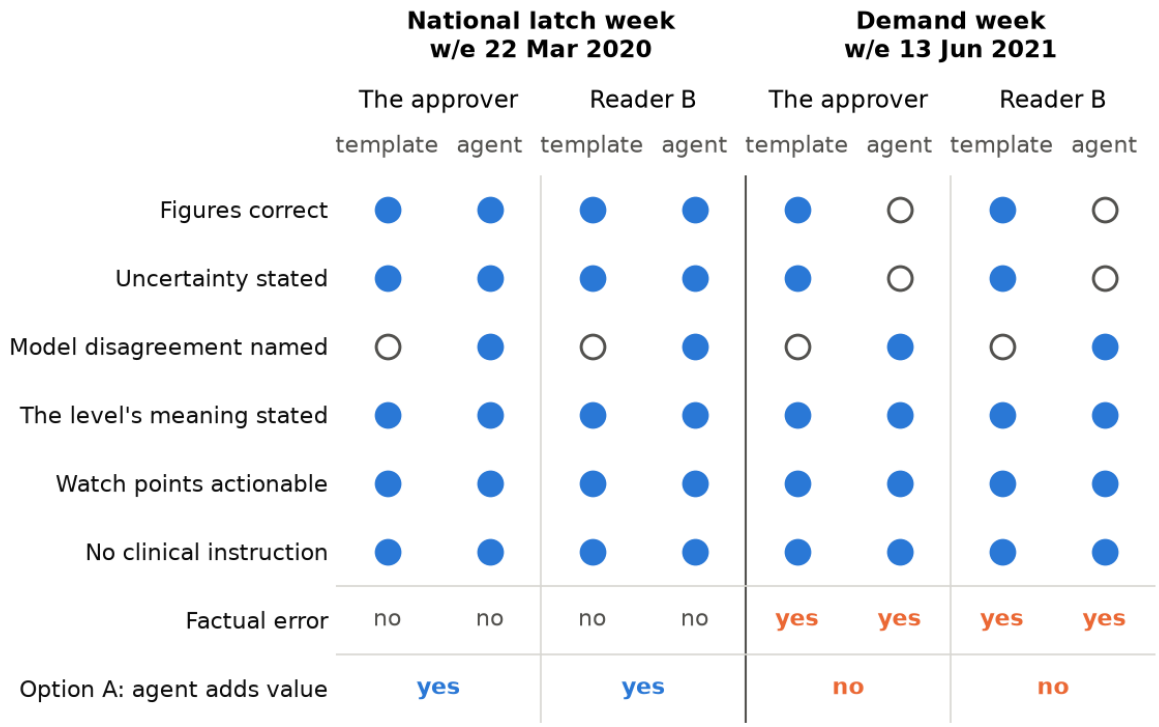
In-sample. The mean of 2019, 2023, 2024 and 2025, whose weeks given notice range from 42 to 71. A busy week is given notice when any flagged unlatched outlook covered it.

What did not work. The adequacy test, stated in advance, needed 6 of the 7 replay briefs verified and meeting at least 5 of the 6 criteria. Neither model passed: the frontier model came closest, at 5 of 7, and the approver chose its briefs. "Briefs are useful to a person" (G6) had two tests. First, 6 of the 7 briefs had to be approved with no factual change; 5 were, as the Reviewer refused both demand-week briefs and the approver rejected them. Second, Reader B had to make the approver's decision on at least 3 of 4 briefs, and did on all 4. So G6 is not met and G7 narrowly missed. Table 3 lists each failure.

The template comparison set the agent's brief against the template brief, fixed text code fills from the same tool results. Judged by a rule fixed before the ratings (Figure 8), the agent's explanation added value in the national latch week for both raters.

Figure 8

Template Against Agent: Both Raters, Both Weeks



Filled: the rater marked the criterion met; open: not met. Option A counts a factual error in the agent's text only. Neither pair was blind to Reader B: each agent text was one of the four briefs Reader B had rated, and the demand week's carried its NOT VERIFIED banner. The template was shown first, as Text A, in both weeks. The approver's marks are not blind.

Note. Option A: the agent's text adds value in a week only if it meets every criterion the template's text meets, and at least one more, with no factual error. Raters marked the demand-week agent text's figures wrong because of its NOT VERIFIED banner, though they passed the code check.

Only the agent's text named the models' disagreement, by marking every model untrusted while latched. In the demand week it did not add value: that text was the refused brief, whose trend and fit claims the checks had stopped. So overall it adds value for neither rater.

Table 3

Failure Cases, With Result and Cause

Run or case	Result	Cause
Local model, round 1	3 of 7 completed before its 840-second budget stopped it; the code check refused every draft, so none was written or verified	A fault in my code that calls the model sent each retry blind, without the first attempt's problems; fixed before round 2
Local model, round 2	7 of 7 completed; none written or verified; a retry with thinking off is untried	Every draft was cut at the 8,192-token output cap, thinking included
Frontier model, round 2	5 of 7 verified; both demand-week briefs refused, then rejected; the template now writes demand weeks	Ninewells' claimed a rising trend from one figure, Lorn & Islands' a flat forecast from one; both echoed code's fit line
"Raised levels are rare in steady years" (G7), under 10%	10.4% to 11.7% in 2018, 2023 and 2024; by design no raised level sets staffing without a manager's approval	My untested reading: demand outgrew the years that set capacity; shadow mode tests it
COVID-19, March 2020	Every model missed, one by 152.8%; the latch kept forecasts untrusted	A shock breaks any model trained on past weeks; shadow mode tests the latch
No notice of 85.5 of 142.5 busy hospital-weeks in the mean steady year	By design the system only adds information, so those weeks rest on the existing process, as before	A rare flag trades recall for precision; the pilot learns what planners use

Note. Replay briefs are seven real past weeks; the notebook and case study give each in full.

By the rule fixed before the ratings, the template now explains demand weeks by default and the agent latch, surge and model-disagreement weeks, though the last has no threshold yet.

Template briefs are queued for approval like the agent's. The shipped template, not the one rated, drops the fit line both raters marked wrong and uses Figure 5's layout, so the pilot tests both.

An AI independent judge read the seven briefs first and drafted the five edits the approver adopted, none of them factual. An AI reviewer, not the system's Reviewer, scored the adequacy test. I built the system and played the approver, which the second reader exists to check.

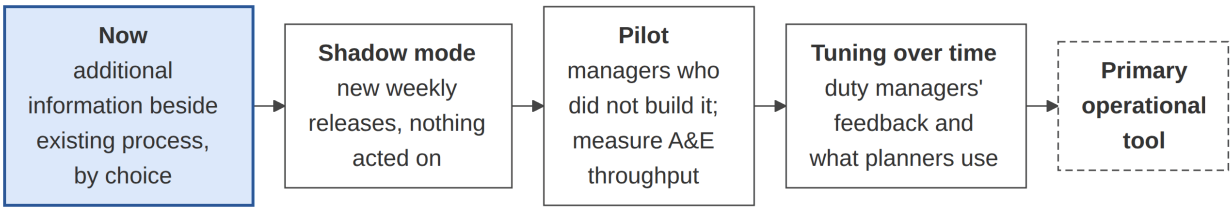
What I learned. On performance and complexity, the GRU's refit and seeds bought no measurable gain, and the frontier model costs money per call and relies on an outside service. On responsible use, integration's value lay in the checks between parts, which stopped both unsupported briefs. The agent explained latch and surge weeks well enough to approve, but overreached in both demand weeks.

Professional Relevance

I kept this a pilot build on purpose: Applied Machine Learning's first release left its payback for a pilot to show, and so does this one. Figure 9 shows the path to a primary operational tool.

Figure 9

From the Pilot Build to a Primary Operational Tool



This work prepares me for AI solutions architect and machine learning engineer roles, drawing on my enterprise resource planning (ERP) work, where people approve a forecast before anyone acts. Its habits are what I would bring to an NHS AI team: tests stated in advance, a person at every decision, and each failure reported with its cause.

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Glossary: every term used in this paper is defined in glossary.pdf, submitted with it.